



SGNP Bird Monitoring Programme

Annual Report

2021/2022



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FOREWORD

The Sanjay Gandhi National Park (SGNP) and its surrounding area form a unique cityscape for Mumbai. It is possibly the largest national park located within the environs of a metropolitan city and is spread across almost 20% area of the city.

Urban forests play an important role in the environmental conditions of a city as they create their micro-climate and moderate the climate of that city. The National Park has a footfall of over a million visitors a year and attracts many nature lovers, including bird watchers! It is one of the most researched areas in Maharashtra. In 2014, an attempt to document the diversity of birds was made with the help of citizen volunteers and a report is available. Long-term monitoring of avian diversity of SGNP using scientific methods and citizen science was believed necessary, considering its proximity to a metropolis and to understand the population dynamics of bird species of the Park.

In February 2021, BNHS, began monitoring the birds in Sanjay Gandhi National Park and Tungreshwar Wildlife Sanctuary. It urged its members to participate in this monthly monitoring, which was done along 12 determined transects. We documented the data every month with the help of the Programmes Department and CEC-Mumbai along with over 500 BNHS members, volunteers, and SGNP Forest Guards. Despite a few setbacks due to the COVID lockdowns, the first year's data was collected and is presented in this report.

A citizen science project like this encourages similar projects for other protected and non-protected areas, around the country. Thus, allowing more data to be collected and helping to understand our avian biodiversity and further efforts to mitigate avian diversity conservation.

I take this opportunity to thank the management of SGNP, especially Sri Mallikarjuna (Director, SGNP) and the staff of the forest department, BNHS members who volunteered, and the devoted team of leaders and citizens for making this bird monitoring a success!

Bivash Pandav, Ph.D.

Director, BNHS

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1. Introduction

Sanjay Gandhi National Park, a part of the northern Western Ghats – one of the richest biodiversity hotspots of the world – is situated in the heart of a metropolitan city and is well known for its biodiversity. It is surrounded by the city of Mumbai & Thane forming an oasis of preserved wildlife. The Park lies between 72°53` to 72°58`E & 19°08` to 19°21` N and covers an area of approx. 103 sq. km and is to the north of the city. The Park has a variety of habitats and is therefore species-rich. Both the geology and vegetation of SGNP are typical of the northern Western Ghats region of the Indian Subcontinent. The Park has, so far, 40 species of mammals including big cats like the Leopard, 251 species of birds, 170 species of butterflies, 38 species of reptiles, 9 species of amphibians, 61 species of Odonata, and more than 1,000 species of plants (Pandya & Sawant, 2014).

Although the Park is a popular destination for many birdwatchers and wildlife enthusiasts, regular bird monitoring for comparative studies to determine the avian diversity, –considering the various pressures it struggles with due to its unique location– has not been done. Therefore, there was a need for a comprehensive bird survey in this protected area.

Dr Parvish Pandya and Mr Saurabh Sawant designed and executed a citizen science project in January 2014 for the Park with the support of the SGNP, Maharashtra Forest Department.

Pandya & Sawant (2014) provided an inventory of species, abundance and movement status, IUCN Redlist status, foraging guilds and habitat association, and a brief on the sampling sites with general information on habitats. The final report of the year-long survey (2014–2015) also provided an additional list of validated species recorded by the authors and other birdwatchers in the recent past in SGNP along with old records from a detailed list compiled by the late Mr Humayun Abdulali in 1981, and for the surrounding area.

The current survey plans to include additional areas in SGNP and Tungarewar Wildlife Sanctuary and hopes to add to this existing data over the next five years to better understand the avian diversity and movement helping us better develop a long-term conservation plan.



The first monthly Bird Monitoring was organized on 28 February 2021. From left to right: Mr. Sanjay Kamble (ACF, SGNP), Ms Dipti Humraskar, Dr Bivash Pandav (Director, BNHS), Mr. Mallikarjuna (Director, SGNP, IFS), Mr Debi Goenka (Hon. Secretary, BNHS), Dr. Parvish Pandya (BNHS Life member), Mr Saurabh Sawant, and Dr Raju Kasambe (Asst. Director, BNHS). Bombay Natural History Society (BNHS) and Sanjay Gandhi National Park, Mumbai (SGNP) signed an MoU on 16 February 2021 for starting a long-term monitoring of bird populations in SGNP.



BNHS Members and Volunteers after the first Bird Count held on 28th February 2021

1.1 Objectives

The bird count will be conducted for five years between February 2021 to January 2026, during which the count is planned to be conducted every month.

The broad objectives of the study are to collect baseline ornithological data in Sanjay Gandhi National Park

1. Documentation of the diversity and abundance of birds in Sanjay Gandhi National Park & Tungreshwar Wildlife Sanctuary
2. Get insight into trends in the population dynamics of birds in SGNP through long-term monitoring using scientific methodology.
3. Identifying the status of a Red-listed and/or endemic species recorded.
4. Make the findings available in the public domain in a simplified manner.

Objectives for Volunteer Training

1. Use scientific data collection methods for flexibility of data analysis and improved data sharing capability
2. Focus on documenting avian community composition (species occurrence), and changes in composition, with a secondary goal of documenting abundance, rather than a strict focus on density.
3. Provide methods that can be completed year-round at a site, adding more intensive surveys, as more volunteers and more skilled birders become available through training.
4. Engage more birders. Simplify design. Simplify instructions. Record all birds. Record all observations.

This report is for the first year of the project (Feb 2021 to Dec 2021).

1.2 Study Area

Sanjay Gandhi National Park is situated between 72°53` to 72°58`E & 19°08` to 19°21` N covering an area of 103.09 sq. km. It is divided into two blocks viz. northern and southern by the Bassein creek, also referred to as Vasai Creek.

The northern block of the National Park is called Nagla Block, and forms a corridor between the Tungareshwar Wildlife Sanctuary and the Southern section of the National Park.

The Southern Section is divided into three ranges, Krishnagiri Upavan, SGNP, and Yeoor. Krishnagiri Upavan is a recreational area and is the smallest range in the Park and covers an area of approximately 5.88 km². The SGNP and Yeoor ranges cover a much larger area of approximately 38.57 km² and 58.64 km², respectively. The Park can also be divided into the Borivali and Thane sectors. The Borivali sector is c. 44.45 km² and the Thane sector is c. 58.64 km² (Bhale *et al.* 2004).

There are also two lakes in the Park, Tulsi and Vihar, which cover an area of approximately 8.5 km². The elevation ranges between 30 m and 466 m above sea level from the creek region to Jambhul Maal the highest point in SGNP (Paranjpye, 1997).

Towards the south, Haathi Gate (Elephant gate / CEC / Film city) forms the southwestern boundary of the Park.

The Tungareshwar Wildlife Sanctuary is spread across 85 sq. km. It forms a wildlife corridor between SGNP and Tansa Wildlife Sanctuary.

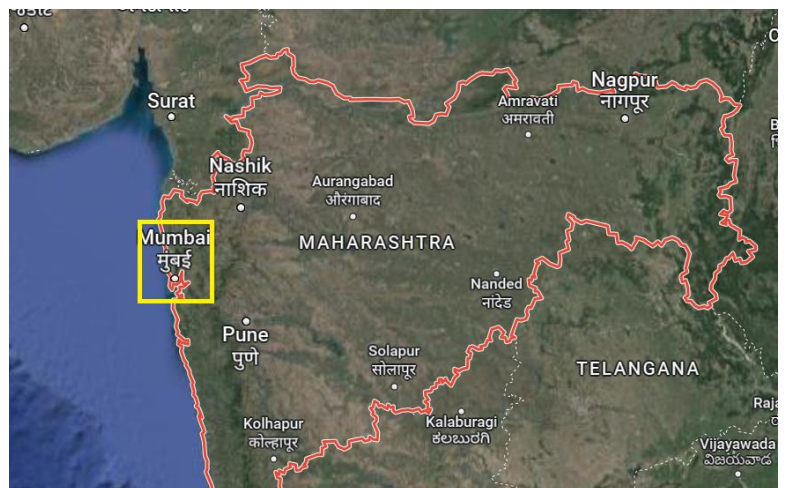
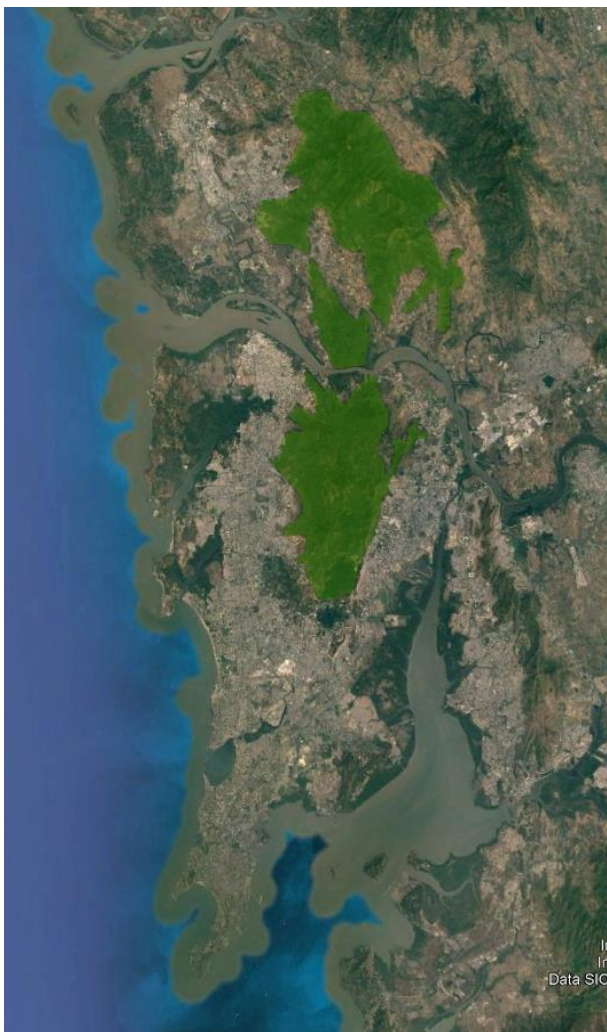
Ten locations throughout SGNP and two locations in Tungareshwar Wildlife Sanctuary, which is located on the eastern plateau of Vasai in Palghar district (previously Thane district) were selected and sampled during 2021, covering areas with different habitat types. Table 1 gives the list of the transects.

Table 1: List of Transects for SGNP Bird Count

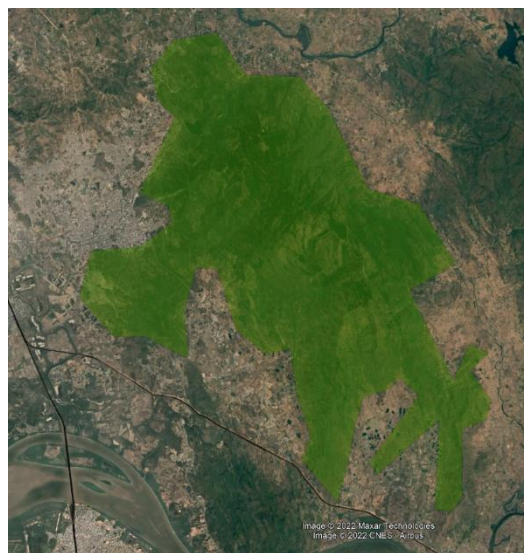
Transect No.	Sector/ Range	Transect
1	Tulsi Sector	Haathi Gate to Culvert 20
2		Culvert 20 to Tulsi Dam
3		Tulshi Dam to Tulsi Gate
4		Thakurpada to Bhoot Bunglow
5	SGNP- Main Sector	Phansacha Paani
6		Jambhul Maal (highest point)
7		Shilonda Trail
8	Nagla Sector	Nagla Gate to Creek
9	Tungareshwar Sector (Wildlife Sanctuary)	Tungareshwar Entrance
10		Tungareshwar Temple
11	Yoor Sector	Yoor Patona Pada to Bhindi Nala
12		Yoor Putranche Bungalows to Gaul Kada

1.2.1 Maps

Maps 1 & 2 Show the study area with respect to Maharashtra and Mumbai-Thane region.

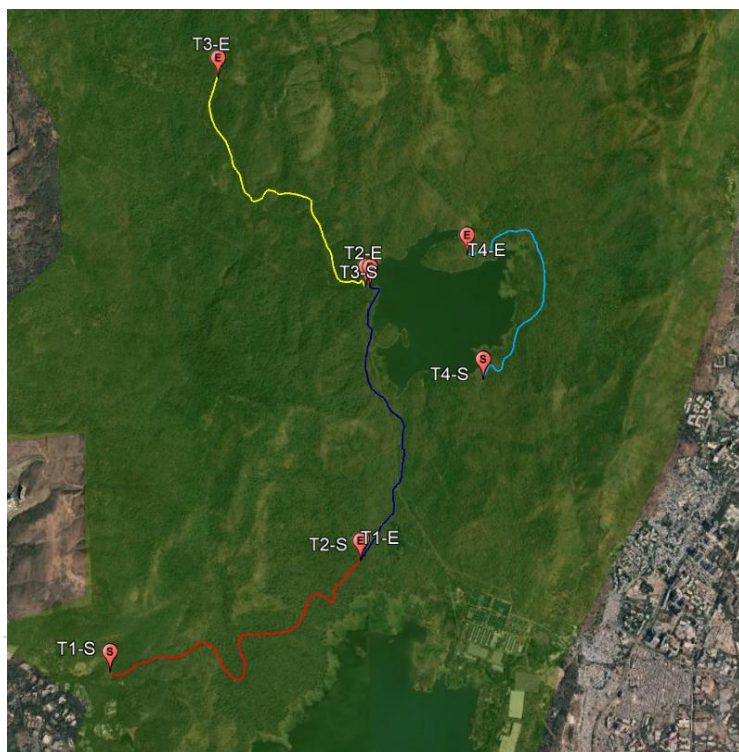


The study area spans two protected areas namely the Sanjay Gandhi National Park (Map 3) and the Tungareshwar Wildlife Sanctuary.



This is further divided into Tulsi Sector (T1, T2, T3, T4), SGNP- Main Sector (T5, T6, T7), Nagla Sector (T8), Tungareshwar WLS Sector (T9, T10), and Thane/Yeoor Sector (T11, T12).

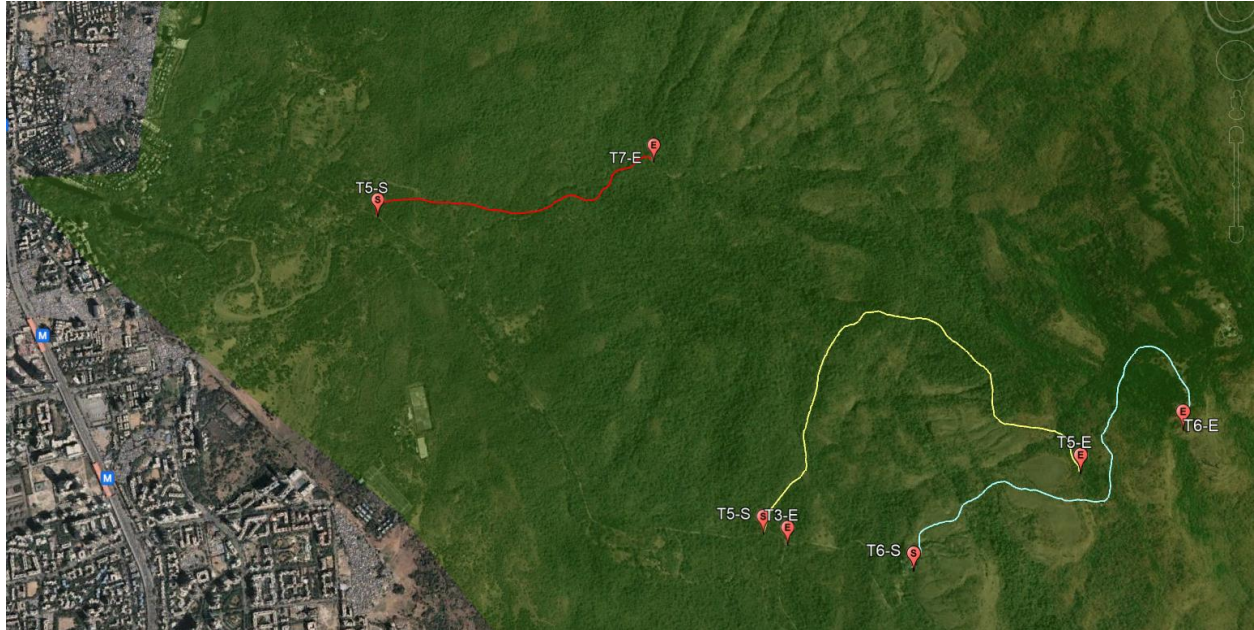
The Tulsi Sector is situated toward the south of the park and consists of four transects (T1, T2, T3 & T4). The T1 starts from the Haathi Gate and continues till Culvery 20. The T2 starts at Culvery 20 and skirts-around Tulsi Lake and ends at the Tulsi dam. T3 at the Tulsi dam and ends at the Kanheri check post gate. T4 starts at the Thakurpada near south bank of the Tulsi Lake and ends at the *Bhooth Banga* situated in the North of the lake atop a small hillock approximately 160 m asl.



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The SGNP Main or the Borivali sector situated in the central area of the main park consist of Three transects (T5, T6 & T7). The T7 transect uses the Shilonda Trail, a popular trail in the park with limited access to the public. While T6 & T7 are situated closer to the central highland of the park. T6 starts at near the Kanheri check post and ends at *Phansacha pani* (310m asl). While T7 starts at the Kanheri Cave ticket counter and ends at the highest point of the SGNP at 468 m asl. Climbing through the eastern slope of the mountain



The Nagla sector covers the Nagla Block of the SGNP which is a 18 sq km of the SGNP that is separated by the Vasai (Bassein) Creek. The area has only one transect, T8 starts at the gate of Nagla in Sasupada travelling towards the old NIC and continuing to end at the creek.

The Thane/Yeoor Sector is situated in the North of the main SGNP and has two transects (T11 & T12), T11 which starts at Patonapada and end at the Bhindi Nala and T12 which starts at the Putrancha Bunglow near the Ghodbunder Road and ends at the *Gaul Kada* (360m asl).



The Tungareshwar sector of the Tungarewar Wildlife Sanctuary, though not a part of the SGNP plays in forming the corridor of protected arear between SGNP and Tansa. Connected to the SGNP through Nagla this sector has two transects (T9 & T10) along the Tungareshwar Rd which connects the NH 48 part of the Golden Quadrilateral Highway at Sativli to the Vajreswari Rd along the Tansa River near Usgaon. The T9 starts at the Tungareshwar entrance and ends at the Tungareshwar Temple, where the T10 begins which climbs and ends at View Point on the Tungarewar Rd.

1.3 Climate

SGNP is located in the north of the western coast of India and the climate is tropical. The temperature ranges from 41°C to 12°C. With May showing the average highest temperature of 33.5 °C and December showing the average lowest temperature of 20.9 °C. The monsoons usually occur between June to September with the mean annual rainfall is approximately 2,600 millimeters during an average period of 87 days per year. (Bhale *et al.* 2004) (Indian Meteorological Department).

1.4 Vegetation

Four major types of plant communities are found in the park. The floral diversity includes over 1,000 species of plants, fungi, and various orchids. The park also has a wide variety of bamboo. One of the amazing aspects of the National Park is that it is constantly undergoing vegetational change. (Paranjpye, 1997). Most part of SGNP has Southern moist teak bearing forest & Southern moist mixed deciduous forest, broadly mixed deciduous forest. Therefore, although the vegetation in SGNP is extremely lush during and just after the monsoons, it becomes quite dry and sparse during the summer months. (Bhale *et al.* 2004) Nagla region boasts of large mangrove forest and scrub along with the semi-evergreen type of forest along the trail. Predominant trees being Teak (*Tectona grandis*), *Grewia sp.*, *Albizia procera*, *Garuga pinnata*, *Careya arborea*, *Morinda tinctoria*, *Oroxylum indicum*, *Holarrhena antidysenterica*, with seasonal bursts of colour by the Silk Cotton Tree (*Bombax ceiba* and *Salmalia insignis*), *Butea monosperma*, the Coral Tree (*Erythrina indica*), *Sterculia colorata* and the leaves of *Schleichera oleosa*. Other striking items of vegetation are the climbers *Combretum ovalifolium* and *Calycopteris floribunda*, patches of bamboo, the 7- or 8-year flowering cycle of the Karvi (*Carvia* = *Strobilanthes callosa*) and the large number of monsoon shrubs including the Glory Lily (*Gloriosa superba*) the Balsam (*Impatiens balsamina*) *Costus speciosus*, *Helicteres isora* and many other brightly coloured flowers. A good number of flowering/nectar plants and fruit bearing trees can be found across the park, which promises an extremely good habitat for many organisms (Pandya & Sawant, 2014).

1.5 Fauna

SGNP boasts of having high biodiversity, especially considering that it is surrounded by cities and is only 103.09 km². It has approximately 250 species of birds. On the whole, the region boasts of having 95 percent of all the raptor species found in India. There are around 40 species of mammals, including about 8 rare or endangered species and approximately 17 species of bats, such as the

Indian Flying Fox (*Pteropus giganteus*). There are also over 40 species of herpetofauna along with a plethora of invertebrates. (p.14, management plan 2001-2002 to 2010-2011 & Bhale *et al.* 2004) An innumerable number of insects can be found in SGNP and it has been estimated that there are more than 8,000 different species. SGNP's Bassein Creek is an ideal habitat for many species, including approximately 24 species of marine fish, making it almost 50 species of fish including freshwater fish (Pandya & Sawant, 2014).

2 Methodology

A study was designed for the survey by using line transects distance sampling method. The bird count is conducted every month. As this is a volunteer based / citizen science project, interested & seasoned birdwatchers and willing volunteers from the BNHS Members of the Mumbai-Thane region were shortlisted, screened and briefed by the core team of consultants coordinating this project. Volunteers with minimum requirements were carefully sorted and chosen to be part of the survey. A total of twelve teams led by 12 experts consisting of two volunteers and a forest guard were formulated. The number of individuals per team was kept low to ensure minimum impact on the data collected. Every month since February 2021 the transects were sampled, each team collected data on the ebird app by making a checklist every 15 mins, while walking the transect. A total of 8 such lists are made for each transect covering a total distance of 3 km. During the reading information like the species along with the number of individuals is collected. During the 2021-22 a total of 9 readings were taken as mentioned in table no 2. The readings for 3 months were missed due to COVID pandemic (and lockdown) and the one for July had to be rescheduled for 1st of August on account of rains.

The birds were observed with binoculars (10x50, 8x42 or 8x40) and identified with the help of field guides of Ali and Ripley (1987), Grimmet *et al.* (1998), Kazmierczak (2000) and Rasmussen & Anderton (2005).

The scientific and common names used in this article follow the BirdLife International bird checklist available online, albeit with a style change in the common name, for which Ali & Ripley (1987), and Manakadan & Khan (2020) were followed.

Table 2: Details of Bird Counts Conducted from February 2021 to January 2023:

Month	Date
February	28/02/2021
March	21/03/2021
April	Survey Not Conducted
May	Survey Not Conducted
June	27/06/2021
July	1/8/2021
August	29/08/2021
September	26/09/2021
October	17/10/2021
November	21/11/2021
December	26/12/2021
January	Survey Not Conducted

3 Results and Discussion

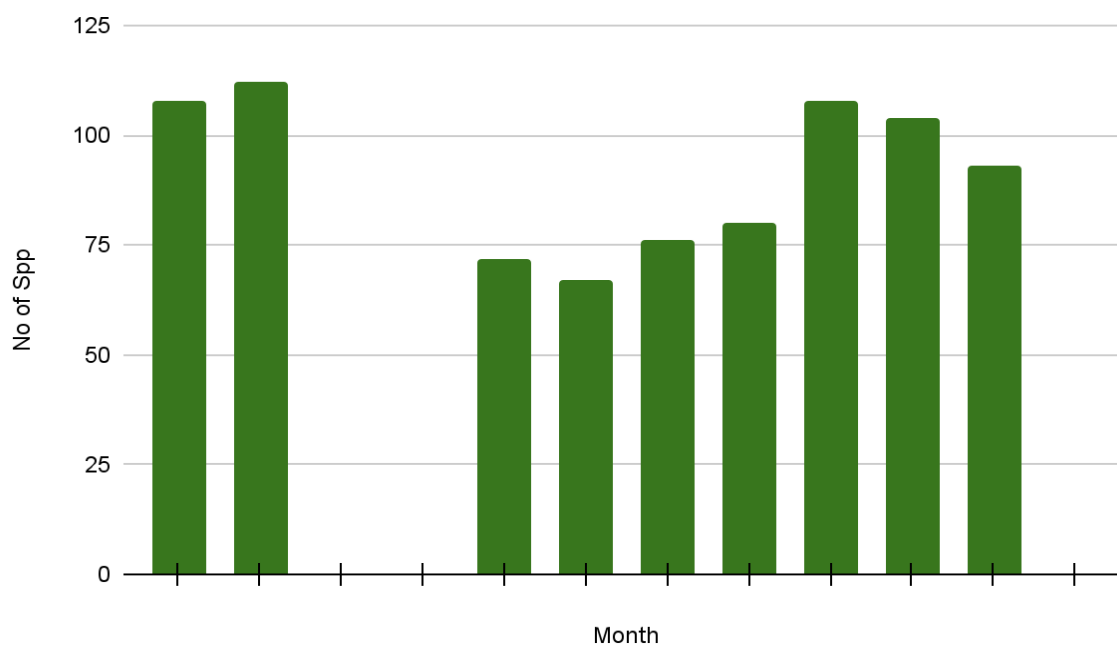
A total of 12 locations were sampled 9 times (see table. 1) during the duration of the survey from 28th February 2021 to January 2023. Total 12 transects were walked. An impressive **186 species belonging to 59 families** were recorded including some **records of red listed species birds**. A maximum number of species were recorded from the Jambhul Maal Transect with 88 species, the number of species for each transect is given in table 3 and chart 1. A total of 130 residents; 45 winter migrants; 2 summer migrants and 10 passage migrants were recorded. A complete checklist has been provided in **(Appendix 1)** along with this an additional annotated list of IUCN Red Listed, endemic and rare species has been given below.

Table 3: Number of bird species recorded during monthly monitoring (February 2021 to January 2022):

Month	Species recorded
February	108
March	112
April*	NA
May*	NA
June	72
July	67
August	76
September	80
October	108
November	104
December	93
January*	NA

**NA: Monitoring not done due to COVID-19 restrictions.*

Chart 1: Total number of bird species recorded in each month



3.1 Species Representation:

3.1.1 Familywise Distribution

The 186 species recorded during the year belong to 59 families. Accipitridae and Muscicapidae showed the most representation with 14 species recorded in each family. While Cuculidae, Ardeidae and Picidae were the next most represented families with 8 species in each family. A complete list of the families and the number of species recorded in each family is given in table 4.

Table 4: Familywise abundance of bird species recorded during February 2021 to January 2022

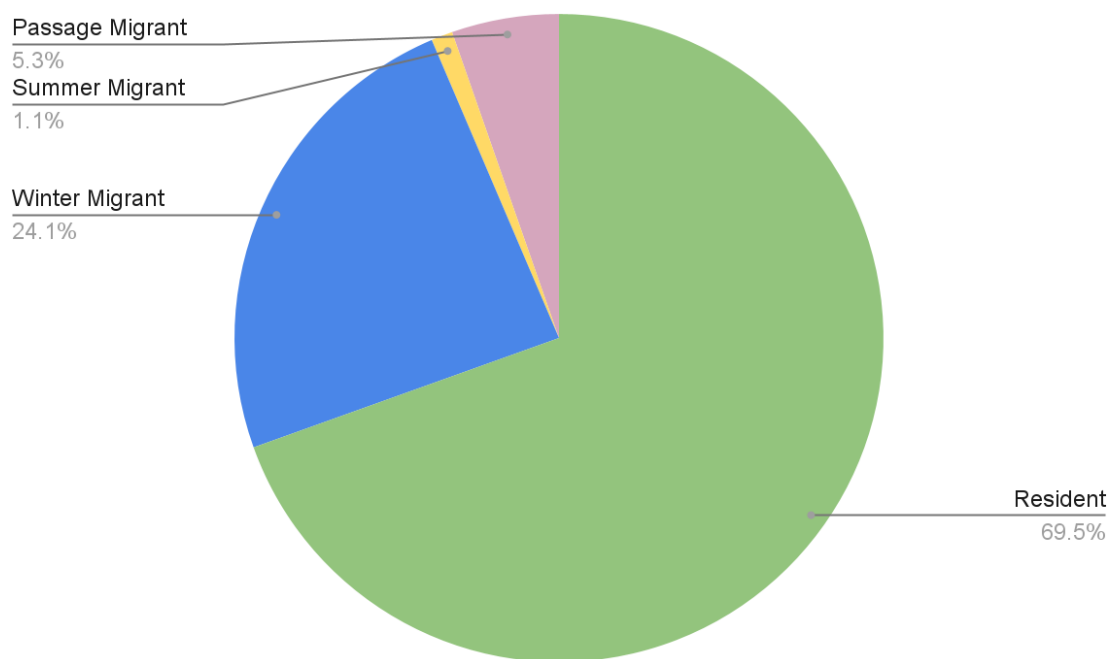
Sr	Family	No of sp		Sr	Family	No of sp
1	Anatidae	2		31	Rhipiduridae	1
2	Phasianidae	4		32	Dicruridae	6
3	Podicipedidae	1		33	Monarchidae	2
4	Columbidae	7		34	Laniidae	1
5	Cuculidae	8		35	Corvidae	3
6	Apodidae	3		36	Stenostiridae	1
7	Rallidae	1		37	Cisticolidae	5
8	Charadriidae	1		38	Acrocephalidae	4
9	Scolopacidae	3		39	Hirundinidae	3
10	Laridae	3		40	Pycnonotidae	4
11	Ciconiidae	2		41	Phylloscopidae	7
12	Phalacrocoracidae	2		42	Sylviidae	1
13	Ardeidae	8		43	Paradoxornithidae	1
14	Threskiornithidae	2		44	Zosteropidae	1
15	Pandionidae	1		45	Timaliidae	2
16	Accipitridae	14		46	Pellorneidae	1
17	Strigidae	5		47	Alcippeidae	1
18	Upupidae	1		48	Leiothrichidae	2
19	Bucerotidae	1		49	Sturnidae	4
20	Alcedinidae	3		50	Turdidae	2

21	Meropidae	2		51	Muscicapidae	14
22	Coraciidae	1		52	Dicaeidae	2
23	Megalaimidae	2		53	Nectariniidae	5
24	Picidae	8		54	Chloropseidae	2
25	Falconidae	3		55	Estrildidae	2
26	Psittacidae	4		56	Passeridae	2
27	Campephagidae	3		57	Motacillidae	5
28	Oriolidae	3		58	Fringillidae	1
29	Vangidae	1		59	Emberizidae	2
30	Aegithinidae	1				

3.1.2: Species Movement:

Also, a total of 130 residents; 44 winter migrants; 3 summer migrants and 9 passage migrants were recorded, as represented in the Chart 2.

Chart 2: Movement of Species



3.2 Significant sightings:

These birds have been considered to be significant because of their global status or status in surrounding regions and available information. Detailed description has been provided along.

3.2.1. River Tern *Sterna aurantia* VU

The River Tern is categorized as Vulnerable on the IUCN Red list. The Species been uplisted to Vulnerable from Near Threatened (2016) on the basis that increasing human disturbance, dam construction, waterbody management, and predation are causing a rapid population decline (IUCN, 2021). Four individuals were recorded on the Nagla Transsect on 21st November 2021.

3.2.2. Indian Spotted Eagle *Clanga hastata* VU

The Indian Spotted Eagle is categorized as Vulnerable on the IUCN Red list. This widespread but poorly known species is thought to have a small and declining population. It therefore qualifies as Vulnerable (IUCN, 2021).

Single individuals were spotted on four occasions at the following transects: Jambhul Maal, Haathi Gate to Culvert 20 and Patonapada to Bhendi Nala Transect.

3.2.3. Greater Spotted Eagle *Clanga clanga* VU

The Greater Spotted Eagle is categorized as Vulnerable on the IUCN Red list. This species has a small population which is suspected to be declining at a rate of >30% over three generations as a result of extensive habitat loss and persistent persecution (IUCN, 2021).

The Greater Spotted Eagle has been spotted on five occasions on the following transects- Jambhul Maal, Haathi Gate to Culvert 20, Phansachapani to Kanheri, Culvert 20 to Tulsi Dam, and Patonapada to Bhendi Nala transects.

3.2.4. Steppe Eagle *Aquila nipalensis* EN

The Steppe Eagle is categorized as Endangered on the IUCN Red list. Though the species outside Europe where it was not thought to be declining at a sufficiently rapid rate to approach the threshold for Vulnerable. Recent information suggests that the population outside Europe may be exposed to greater threats than was previously thought and has also undergone very rapid recent declines across much of the range. It is therefore classified as Endangered (IUCN, 2021).

Two individuals were spotted at the Jambhul Maal transect and a single individual was at the Bhoot Banglow Transect on 26th Dec 2021.

3.2.5. Painted Stork *Mycteria leucocephala* NT

The Painted Stork is categorized as Near Threatened on the Red list. Although one of the most abundant of the Asian storks, this species is classified as Near Threatened because it is thought to be undergoing a moderately rapid population decline owing primarily to hunting, wetland drainage and pollution (IUCN, 2021).

Five individuals were spotted on 27th June 2021 and a single individual on 26th Sept at the Nagla transects and a single individual spotted on 28th February 2021 at the Haathi Gate to Culvert 20 transect.

3.2.6. Black-headed Ibis *Threskiornis melanocephalus* NT

The Black-headed Ibis is categorized as Near Threatened on the Red list. In common with most large wetland species in Asia, this species is thought to be undergoing a population reduction, which is suspected to be moderately rapid (IUCN, 2021).

A single individual was spotted on 21st Mar 2021 at the Haathi Gate to Culvert 20 transect.

3.2.7 Alexandrine Parakeet *Palaeornis eupatria* NT

The Alexandrine Parakeet is categorized as Near Threatened on the IUCN Red list. Although it remains common in some areas and the status is clouded by feral/escaped populations, it is suspected to be undergoing a moderately rapid population decline owing to on-going habitat loss, persecution and trapping pressure. (IUCN, 2021)

The species is fairly common, spotted on 7 occasions. Haathi Gate to Culvert 20 transects showed the most frequency of the species.

3.3 List of Species

A total of 186 species were recorded during the period, a list of species with the following details like the Common Name, Scientific name, Status and IUCN Red list Category has been given in Annexure.

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5 Acknowledgments

5.1 Core Team

Dr Raju Kasambe, Asif N. Khan, Dipti Humraskar, Dr Parvish Pandya, and Saurabh Sawant.

5.2 Team of BNHS Staff

Abhijit Malekar, Kiran Thumma, Nandkishor Dudhe, Niketan Thakur, Prathemesh Joshi, Priya James, Priyanka Jundare, Sagar Mahajan and Viraj Athalye.

5.3 Team Leaders

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Alok Bhave	Jayakrishnan Natarajan	Ritvika Patodia
Alok Bhave	Jayant Barve	Ronit Datta
Amit Goenka	Kalpesh Gaitonde	Salil Choksi
Amogh Wagh	Kalyani K Kolvankar	Samir Eranezhath
Amol Lopes	Ketki Marthak, Dr	Sanjeev Deo
Anand Kaundanya	Krupalu Gharekhan	Sanket Mhatre
Anoushka Datta	Mark D'Mello	Shainaz Jussa
Aparna Rao	Naman Kaji	Shannon Pereira
Ashvini Menon	Neha Savla	Shreya Sahu
Awadh Agrawal	Nishant Bhagwat	Shubham Soni
Beena Menon	Nosherwan Sethna	Snehal Gaikwad
Debangana Chakravorty	Prachi Galange	Varun Satose
Deeip Dicondwar	Prakash Goray	Vedant Kasambe
Deepak Gujar	Pravin Subramaniam	Vidhin R Gandhi
Dheeraj Nahata	Rahul Pereira	Viral Mistry
Francis D'Souza	Rajesh Dubey	Vivek Pande
Gajanan Karanjikar Capt.	Ramesh Shenai	Yogesh Patel

Annexure

Checklist of 186 species that were recorded during the period 2021-2022 of the SGNP Bird Monitoring Programme, with the following details like the Common Name, Scientific name, Status and IUCN Red list Category

Sr	Common Name	Sci. name	Family	Status	IUCN
1	Indian Spot-billed Duck	<i>Anas poecilorhyncha</i>	Anatidae	R	LC
2	Green-winged Teal	<i>Anas crecca</i>	Anatidae	W	LC
3	Indian Peafowl	<i>Pavo cristatus</i>	Phasianidae	R	LC
4	Red Spurfowl	<i>Galloperdix spadicea</i>	Phasianidae	R	LC
5	Grey Junglefowl	<i>Gallus sonneratii</i>	Phasianidae	R	LC
6	Little Grebe	<i>Tachybaptus ruficollis</i>	Podicipedidae	R	LC
7	Rock Pigeon	<i>Columba livia</i>	Columbidae	R	LC
8	Oriental Turtle-Dove	<i>Streptopelia orientalis</i>	Columbidae	R	LC
9	Eastern Spotted Dove	<i>Spilopelia chinensis</i>	Columbidae	R	LC
10	Laughing Dove	<i>Spilopelia senegalensis</i>	Columbidae	R	LC
11	Asian Emerald Dove	<i>Chalcophaps indica</i>	Columbidae	R	LC
12	Grey-fronted Green-Pigeon	<i>Treron affinis</i>	Columbidae	R	LC
13	Yellow-footed Green-Pigeon	<i>Treron phoenicopterus</i>	Columbidae	R	LC
14	Greater Coucal	<i>Centropus sinensis</i>	Cuculidae	R	LC
15	Pied Cuckoo	<i>Clamator jacobinus</i>	Cuculidae	S	LC
16	Asian Koel	<i>Eudynamys scolopaceus</i>	Cuculidae	R	LC
17	Banded Bay Cuckoo	<i>Cacomantis sonneratii</i>	Cuculidae	R	LC
18	Fork-tailed Drongo-Cuckoo	<i>Surniculus dicruroides</i>	Cuculidae	R	LC
19	Common Hawk-Cuckoo	<i>Hierococcyx varius</i>	Cuculidae	R	LC
20	Indian Cuckoo	<i>Cuculus micropterus</i>	Cuculidae	P	LC
21	Common Cuckoo	<i>Cuculus canorus</i>	Cuculidae	S	LC
22	Alpine Swift	<i>Tachymarptis melba</i>	Apodidae	R	LC
23	Little Swift	<i>Apus affinis</i>	Apodidae	R	LC
24	Asian Palm-Swift	<i>Cypsiurus balasiensis</i>	Apodidae	R	LC
25	White-breasted Waterhen	<i>Amaurornis phoenicurus</i>	Rallidae	R	LC
26	Red-wattled Lapwing	<i>Vanellus indicus</i>	Charadriidae	R	LC
27	Common Sandpiper	<i>Actitis hypoleucos</i>	Scolopacidae	W	LC

28	Green Sandpiper	<i>Tringa ochropus</i>	Scolopacidae	W	LC
29	Common Redshank	<i>Tringa totanus</i>	Scolopacidae	W	LC
30	Gull-billed Tern	<i>Gelochelidon nilotica</i>	Laridae	W	LC
31	Whiskered Tern	<i>Chlidonias hybrida</i>	Laridae	W	LC
32	River Tern	<i>Sterna aurantia</i>	Laridae	R	VU
33	Asian Openbill	<i>Anastomus oscitans</i>	Ciconiidae	W	LC
34	Painted Stork	<i>Mycteria leucocephala</i>	Ciconiidae	R	NT
35	Little Cormorant	<i>Microcarbo niger</i>	Phalacrocoracidae (	R	LC
36	Indian Cormorant	<i>Phalacrocorax fuscicollis</i>	Phalacrocoracidae (	R	LC
37	Grey Heron	<i>Ardea cinerea</i>	Ardeidae	R	LC
38	Purple Heron	<i>Ardea purpurea</i>	Ardeidae	R	LC
39	Great Egret	<i>Ardea alba</i>	Ardeidae	R	LC
40	Intermediate Egret	<i>Ardea intermedia</i>	Ardeidae	R	LC
41	Little Egret	<i>Egretta garzetta</i>	Ardeidae	R	LC
42	Cattle Egret	<i>Bubulcus ibis</i>	Ardeidae	R	LC
43	Indian Pond-Heron	<i>Ardeola grayii</i>	Ardeidae	R	LC
44	Black-crowned Night-Heron	<i>Nycticorax nycticorax</i>	Ardeidae	R	LC
45	Glossy Ibis	<i>Plegadis falcinellus</i>	Threskiornithidae	W	LC
46	Black-headed Ibis	<i>Threskiornis melanocephalus</i>	Threskiornithidae	R	NT
47	Osprey	<i>Pandion haliaetus</i>	Pandionidae	W	LC
48	Black-winged Kite	<i>Elanus caeruleus</i>	Accipitridae	R	LC
49	Oriental Honey-buzzard	<i>Pernis ptilorhynchus</i>	Accipitridae	R	LC
50	Crested Serpent-Eagle	<i>Spilornis cheela</i>	Accipitridae	R	LC
51	Black Eagle	<i>Ictinaetus malaiensis</i>	Accipitridae	R	LC
52	Indian Spotted Eagle	<i>Clanga hastata</i>	Accipitridae	R	VU
53	Greater Spotted Eagle	<i>Clanga clanga</i>	Accipitridae	W	VU
54	Booted Eagle	<i>Hieraaetus pennatus</i>	Accipitridae	W	LC
55	Steppe Eagle	<i>Aquila nipalensis</i>	Accipitridae	W	EN
56	White-eyed Buzzard	<i>Butastur teesa</i>	Accipitridae	R	LC
57	Shikra	<i>Accipiter badius</i>	Accipitridae	R	LC
58	Eurasian Sparrowhawk	<i>Accipiter nisus</i>	Accipitridae	W	LC
59	Black Kite	<i>Milvus migrans</i>	Accipitridae	R	LC
60	Brahminy Kite	<i>Haliastur indus</i>	Accipitridae	R	LC

61	Common Buzzard	<i>Buteo buteo</i>	Accipitridae	W	LC
62	Jungle Owlet	<i>Glaucidium radiatum</i>	Strigidae	R	LC
63	Spotted Owlet	<i>Athene brama</i>	Strigidae	R	LC
64	Mottled Wood-Owl	<i>Strix ocellata</i>	Strigidae	R	LC
65	Short-eared Owl	<i>Asio flammeus</i>	Strigidae	W	LC
66	Brown Hawk-Owl	<i>Ninox scutulata</i>	Strigidae	R	LC
67	Eurasian Hoopoe	<i>Upupa epops</i>	Upupidae	R	LC
68	Indian Grey Hornbill	<i>Ocyeros birostris</i>	Bucerotidae	R	LC
69	Common Kingfisher	<i>Alcedo atthis</i>	Alcedinidae	R	LC
70	Black-backed Dwarf-Kingfisher	<i>Ceyx erithaca</i>	Alcedinidae	S	LC
71	White-throated Kingfisher	<i>Halcyon smyrnensis</i>	Alcedinidae	R	LC
72	Green Bee-eater	<i>Merops orientalis</i>	Meropidae	R	LC
73	Blue-tailed Bee-eater	<i>Merops philippinus</i>	Meropidae	W	LC
74	European Roller	<i>Coracias garrulus</i>	Coraciidae	P	LC
75	Coppersmith Barbet	<i>Psilopogon haemacephalus</i>	Megalaimidae	R	LC
76	Brown-headed Barbet	<i>Psilopogon zeylanicus</i>	Megalaimidae	R	LC
77	Heart-spotted Woodpecker	<i>Hemicircus canente</i>	Picidae	R	LC
78	Brown-capped Pygmy Woodpecker	<i>Yungipicus nanus</i> (<i>Picoides nanus</i>)	Picidae	R	LC
79	Yellow-crowned Woodpecker	<i>Leiopicus mahrattensis</i>	Picidae	R	LC
80	Greater Flameback	<i>Chrysocolaptes guttacristatus</i>	Picidae	R	LC
81	White-naped Woodpecker - <i>Chrysocolaptes festivus</i>	<i>Chrysocolaptes festivus</i>	Picidae	R	LC
82	Rufous Woodpecker	<i>Micropternus brachyurus</i>	Picidae	R	LC
83	Common Flameback	<i>Dinopium javanense</i>	Picidae	P	LC
84	Black-rumped Flameback	<i>Dinopium benghalense</i>	Picidae	R	LC
85	Eurasian Kestrel	<i>Falco tinnunculus</i>	Falconidae	W	LC
86	Amur Falcon	<i>Falco amurensis</i>	Falconidae	P	LC
87	Peregrine Falcon	<i>Falco peregrinus</i>	Falconidae	R	LC
88	Alexandrine Parakeet	<i>Palaeornis eupatria</i>	Psittacidae	R	NT
89	Rose-ringed Parakeet	<i>Alexandrinus krameri</i>	Psittacidae	R	LC
90	Plum-headed Parakeet	<i>Himalayapsitta cyanocephala</i>	Psittacidae	R	LC

91	Vernal Hanging-Parrot	<i>Loriculus vernalis</i>	Psittacidae	R	LC
92	Small Minivet	<i>Pericrocotus cinnamomeus</i>	Campephagidae	R	LC
93	Orange Minivet	<i>Pericrocotus flammeus</i>	Campephagidae	R	LC
94	Large Cuckooshrike	<i>Coracina macei</i>	Campephagidae	R	LC
95	Indian Golden Oriole	<i>Oriolus kundoo</i>	Oriolidae	R	LC
96	Black-naped Oriole	<i>Oriolus chinensis</i>	Oriolidae	W	LC
97	Black-hooded Oriole	<i>Oriolus xanthornus</i>	Oriolidae	R	LC
98	Common Woodshrike	<i>Tephrodornis pondicerianus</i>	Vangidae	R	LC
99	Common Iora	<i>Aegithina tiphia</i>	Aegithinidae	R	LC
100	Spot-breasted Fantail	<i>Rhipidura albogularis</i>	Rhipiduridae	R	LC
101	Black Drongo	<i>Dicrurus macrocercus</i>	Dicruridae	R	LC
102	Ashy Drongo	<i>Dicrurus leucophaeus</i>	Dicruridae	W	LC
103	White-bellied Drongo	<i>Dicrurus caerulescens</i>	Dicruridae	R	LC
104	Bronzed Drongo	<i>Dicrurus aeneus</i>	Dicruridae	R	LC
105	Hair-crested Drongo	<i>Dicrurus hottentottus</i>	Dicruridae	R	LC
106	Greater Racket-tailed Drongo	<i>Dicrurus paradiseus</i>	Dicruridae	R	LC
107	Black-naped Monarch	<i>Hypothymis azurea</i>	Monarchidae	R	LC
108	Indian Paradise-Flycatcher	<i>Terpsiphone paradisi</i>	Monarchidae	R	LC
109	Long-tailed Shrike	<i>Lanius schach</i>	Laniidae	R	LC
110	Rufous Treepie	<i>Dendrocitta vagabunda</i>	Corvidae	R	LC
111	House Crow	<i>Corvus splendens</i>	Corvidae	R	LC
112	Large-billed Crow	<i>Corvus macrorhynchos</i>	Corvidae	R	LC
113	Grey-headed Canary-Flycatcher	<i>Culicicapa ceylonensis</i>	Stenostiridae	W	LC
114	Common Tailorbird	<i>Orthotomus sutorius</i>	Cisticolidae	R	LC
115	Grey-breasted Prinia	<i>Prinia hodgsonii</i>	Cisticolidae	R	LC
116	Jungle Prinia	<i>Prinia sylvatica</i>	Cisticolidae	R	LC
117	Ashy Prinia	<i>Prinia socialis</i>	Cisticolidae	R	LC
118	Plain Prinia	<i>Prinia inornata</i>	Cisticolidae	R	LC
119	Booted Warbler	<i>Iduna caligata</i>	Acrocephalidae	W	LC
120	Sykes's Warbler	<i>Iduna rama</i>	Acrocephalidae	P	LC
121	Blyth's Reed Warbler	<i>Acrocephalus dumetorum</i>	Acrocephalidae	W	LC

122	Clamorous Reed Warbler	<i>Acrocephalus stentoreus</i>	Acrocephalidae	R	LC
123	Dusky Crag-Martin	<i>Ptyonoprogne concolor</i>	Hirundinidae	R	LC
124	Barn Swallow	<i>Hirundo rustica</i>	Hirundinidae	W	LC
125	Red-rumped Swallow	<i>Cecropis daurica</i>	Hirundinidae	R	LC
126	Red-vented Bulbul	<i>Pycnonotus cafer</i>	Pycnonotidae	R	LC
127	Red-whiskered Bulbul	<i>Pycnonotus jocosus</i>	Pycnonotidae	R	LC
128	White-browed Bulbul	<i>Pycnonotus luteolus</i>	Pycnonotidae	R	LC
129	Square-tailed Bulbul	<i>Hypsipetes ganeesa</i>	Pycnonotidae	R	LC
130	Hume's Warbler	<i>Phylloscopus humei</i>	Phylloscopidae	W	LC
131	Sulphur-bellied Warbler	<i>Phylloscopus griseolus</i>	Phylloscopidae	W	LC
132	Tickell's Leaf Warbler	<i>Phylloscopus affinis</i>	Phylloscopidae	P	LC
133	Common Chiffchaff	<i>Phylloscopus collybita</i>	Phylloscopidae	W	LC
134	Green Warbler	<i>Phylloscopus nitidus</i>	Phylloscopidae	P	LC
135	Greenish Warbler	<i>Phylloscopus trochiloides</i>	Phylloscopidae	W	LC
136	Western Crowned Warbler	<i>Phylloscopus occipitalis</i>	Phylloscopidae	P	LC
137	Lesser Whitethroat	<i>Curruca curruca</i>	Sylviidae	W	LC
138	Yellow-eyed Babbler	<i>Chrysomma sinense</i>	Paradoxornithidae	R	LC
139	Indian White-eye	<i>Zosterops palpebrosus</i>	Zosteropidae	R	LC
140	Tawny-bellied Babbler	<i>Dumetia hyperythra</i>	Timaliidae	R	LC
141	Indian Scimitar-Babbler	<i>Pomatorhinus horsfieldii</i>	Timaliidae	R	LC
142	Puff-throated Babbler	<i>Pellorneum ruficeps</i>	Pellorneidae	R	LC
143	Brown-cheeked Fulvetta	<i>Alcippe poioicephala</i>	Alcippeidae	R	LC
144	Jungle Babbler	<i>Argya striata</i>	Leiotrichidae	R	LC
145	Common Babbler	<i>Argya caudata</i>	Leiotrichidae	R	LC
146	Rosy Starling	<i>Pastor roseus</i>	Sturnidae	W	LC
147	Malabar Starling	<i>Sturnia blythii</i>	Sturnidae	R	DD
148	Common Myna	<i>Acridotheres tristis</i>	Sturnidae	R	LC
149	Jungle Myna	<i>Acridotheres fuscus</i>	Sturnidae	R	LC
150	Orange-headed Thrush	<i>Geokichla citrina</i>	Turdidae	R	LC
151	Indian Blackbird	<i>Turdus simillimus</i>	Turdidae	R	LC
152	Asian Brown Flycatcher	<i>Muscicapa dauurica</i>	Muscicapidae	W	LC
153	Brown-breasted Flycatcher	<i>Muscicapa muttui</i>	Muscicapidae	W	LC

154	Indian Robin	<i>Saxicoloides fulicatus</i>	Muscicapidae	R	LC
155	Oriental Magpie-Robin	<i>Copsychus saularis</i>	Muscicapidae	R	LC
156	White-rumped Shama	<i>Kittacincla malabarica</i>	Muscicapidae	R	LC
157	Tickell's Blue Flycatcher	<i>Cyornis tickelliae</i>	Muscicapidae	R	LC
158	Verditer Flycatcher	<i>Eumyias thalassinus</i>	Muscicapidae	W	LC
159	Malabar Whistling-Thrush	<i>Myophonus horsfieldii</i>	Muscicapidae	R	LC
160	Ultramarine Flycatcher	<i>Ficedula superciliaris</i>	Muscicapidae	W	LC
161	Taiga Flycatcher	<i>Ficedula albicilla</i>	Muscicapidae	W	LC
162	Red-breasted Flycatcher	<i>Ficedula parva</i>	Muscicapidae	W	LC
163	Black Redstart	<i>Phoenicurus ochruros</i>	Muscicapidae	W	LC
164	Blue-capped Rock-Thrush	<i>Monticola cinclorhyncha</i>	Muscicapidae	P	LC
165	Blue Rock-Thrush	<i>Monticola solitarius</i>	Muscicapidae	W	LC
166	Thick-billed Flowerpecker	<i>Dicaeum agile</i>	Dicaeidae	R	LC
167	Pale-billed Flowerpecker	<i>Dicaeum erythrorhynchos</i>	Dicaeidae	R	LC
168	Purple-rumped Sunbird	<i>Leptocoma zeylonica</i>	Nectariniidae	R	LC
169	Crimson-backed Sunbird	<i>Leptocoma minima</i>	Nectariniidae	R	LC
170	Purple Sunbird	<i>Cinnyris asiaticus</i>	Nectariniidae	R	LC
171	Loten's Sunbird	<i>Cinnyris lotenius</i>	Nectariniidae	R	LC
172	Vigors's Sunbird	<i>Aethopyga vigorsii</i>	Nectariniidae	R	LC
173	Jerdon's Leafbird	<i>Chloropsis jerdoni</i>	Chloropseidae	R	LC
174	Golden-fronted Leafbird	<i>Chloropsis aurifrons</i>	Chloropseidae	R	LC
175	Scaly-breasted Munia	<i>Lonchura punctulata</i>	Estrildidae	R	LC
176	White-rumped Munia	<i>Lonchura striata</i>	Estrildidae	R	LC
177	House Sparrow	<i>Passer domesticus</i>	Passeridae	R	LC
178	Yellow-throated Sparrow	<i>Gymnoris xanthocollis</i>	Passeridae	R	LC
179	Forest Wagtail	<i>Dendronanthus indicus</i>	Motacillidae	W	LC
180	Grey Wagtail	<i>Motacilla cinerea</i>	Motacillidae	W	LC
181	Western Yellow Wagtail	<i>Motacilla flava</i>	Motacillidae	W	LC
182	Paddyfield Pipit	<i>Anthus rufulus</i>	Motacillidae	R	LC
183	Tree Pipit	<i>Anthus trivialis</i>	Motacillidae	W	LC
184	Common Rosefinch	<i>Carpodacus erythrinus</i>	Fringillidae	W	LC
185	Red-headed Bunting	<i>Emberiza bruniceps</i>	Emberizidae	W	LC
186	Grey-necked Bunting	<i>Emberiza buchanani</i>	Emberizidae	W	LC

BOMBAY NATURAL HISTORY SOCIETY

Founded in 1883 for the study of natural history, the Bombay Natural History Society (BNHS) is now one of the premier research and conservation organisations in the country. The Society publishes a journal, the Journal of the Bombay Natural History Society, devoted to natural history and also has a popular publication, Hornbill, for the layman. It has also published a number of books on wildlife and nature. Its library has a large collection of books and scientific journals on wildlife and the environment. The Society's invaluable collection of bird, mammal, reptile, amphibian and insect specimens has been recognised as a National Heritage Collection.

Membership of the Society is open to individuals and institutions within India and abroad. For more details, please write to:

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